Retail Billing with Multiple Discount Rules

1. Problem statement

Develop a Python program to calculate a retail invoice for multiple products, quantities, taxes, and discounts. Apply a maximum discount limit and generate an itemized bill.

2. Objectives

- Calculate product-wise costs.
- Apply a threshold discount.
- Calculate category-specific taxes.
- Limit the maximum discount.
- Generate an invoice.

3. Assumptions

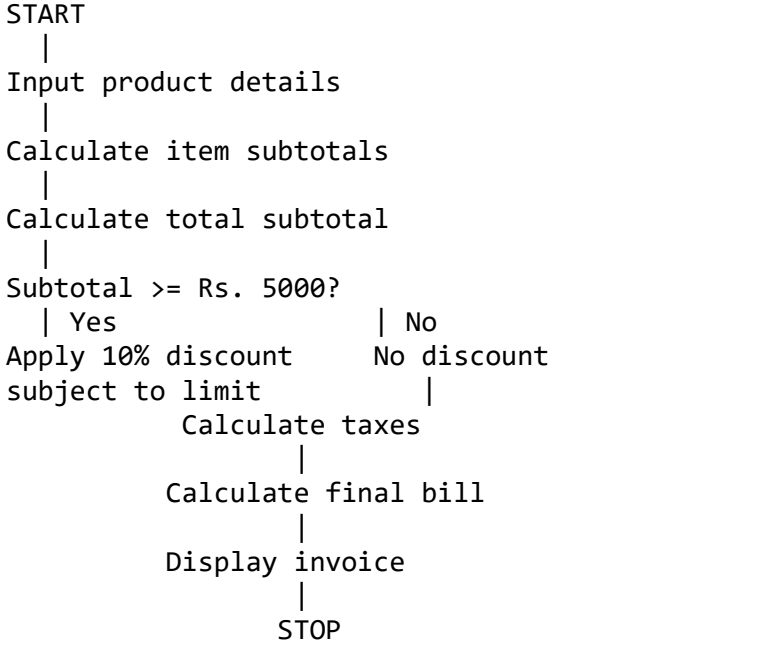
- Electronics tax: 18%.
- Clothing tax: 5%.
- Grocery tax: 0% for this example.
- Discount: 10% if the subtotal is at least ₹5,000.
- Maximum discount: ₹1,000.
- All prices and quantities must be positive.

These tax rates are simplified assignment assumptions, not a statement of applicable real-world tax rates.

4. Algorithm

1. Input the number of products.
2. For each product, input its name, category, price, and quantity.
3. Calculate each item's subtotal.
4. Calculate the total subtotal.
5. Apply the threshold discount, subject to the maximum limit.
6. Calculate taxes on the discounted item amounts.
7. Display the invoice.

5. Flowchart



6. Pseudocode

```
START
INPUT number_of_products
subtotal = 0

FOR each product
    INPUT name, category, price, quantity
    item_total = price * quantity
    ADD item_total to subtotal
END FOR

IF subtotal >= 5000
    discount = MIN(subtotal * 10%, 1000)
ELSE
    discount = 0
END IF

Calculate tax on discounted item amounts
final_bill = subtotal - discount + total_tax
PRINT invoice
STOP
```

7. Python source code

```
n = int(input("Enter number of products: "))

subtotal = 0
products = []

tax_rates = {
    "electronics": 0.18,
    "clothing": 0.05,
    "grocery": 0.00
}

if n <= 0:
    print("Invalid number of products")
else:
    valid = True

    for i in range(n):
        name = input("Product name: ")
        category = input("Category (electronics/clothing/grocery): ").lower()
        price = float(input("Price: "))
        quantity = int(input("Quantity: "))

        if (category not in tax_rates
                or price <= 0 or quantity <= 0):
            valid = False
```

```
            break

        amount = price * quantity
        subtotal += amount
        products.append(
            (name, category, amount)
        )

    if not valid:
        print("Invalid product details")
    else:
        discount = min(subtotal * 0.10, 1000) if subtotal >= 5000 else 0
        ratio = (subtotal - discount) / subtotal
        tax = 0

        print("\n--- Invoice ---")
        for name, category, amount in products:
            discounted_amount = amount * ratio
            item_tax = discounted_amount * tax_rates[category]
            tax += item_tax
            print(name, "Amount:", round(amount, 2),
                  "Tax:", round(item_tax, 2))

        total = subtotal - discount + tax
        print("Subtotal:", round(subtotal, 2))
        print("Discount:", round(discount, 2))
        print("Tax:", round(tax, 2))
        print("Final bill:", round(total, 2))
```

8. Test cases

Test	Products	Subtotal	Expected discount
1	Clothing ₹1,000 × 1	₹1,000	₹0
2	Electronics ₹3,000 × 2	₹6,000	₹600
3	Grocery ₹6,000 × 1	₹6,000	₹600
4	Electronics ₹15,000 × 1	₹15,000	₹1,000